

5.0: Lab 05 Assignment

A special type of situation, called a controlled break, can occur when processing records of data must temporarily pause because a key value has changed. For example, a control break might occur in a report that contains subtotals for groupings of records.

This program requires such a procedure.

Sales people at a local car dealership are paid by commission. For every car sale, a sales person earns 30% of the base price or \$100, whichever is higher.

Write a program that uses of a loop to read through and process records in a file. Whenever the program encounters a new employee id, it should pause processing long enough to display the total of the previous employee's commissions before processing the record just read. After all records have been processed, the program should display a count of the records processed, the total sales, and the total commissions paid out. A sample output follows:

```
WEEKLY SALES REPORT

Employee   Retail      Base   Commission
101        24125.00    1201.00    360.30
101         7650.00     350.00    105.00
101        38460.00    1517.00    455.10
***Total Commission for 101:  920.40

102        10500.00     500.00    150.00
102         7500.00     250.00    100.00
102        17551.00    1120.00    336.00
102        12400.00     400.00    120.00
***Total Commission for 102:  706.00

103         41500.00     550.00    165.00
103        18670.00    1250.00    375.00
103         6700.00     250.00    100.00
103        17067.00    1018.00    305.40
***Total Commission for 103:  945.40

Records:    11
Total Sales: 202123.00
Total Commissions: 2571.80
```

A sample input file, `carsales.dat`, to generate the report is shown below. The first value in a line is the employee id. The second value is the retail sale amount of the car (without tax.) The third item is the base price on which the sales person's commission is determined. Note: It is important that our data set is sorted by sales person id number. If the data set is not sorted, the control-break process won't work.

ID	Retail	Base
101	24125	1201
101	7650	350
101	38460	1517
102	10500	500
102	7500	250
102	17551	1120
102	12400	400
103	41500	550
103	18670	1250
103	6700	250
103	17067	1018

Tips:

- Make sure the input file exists prior to starting to process records. If the file does not exist, end the program.
- When we read the first record, we need to initialize a variable we will use to compare all following records against to see if the employee id has changed. Typically, we complete this step prior to starting the loop which processes all subsequent records.
- During normal processing, we accumulate a subtotal for the current sales person each time we loop. When the key field changes (and the control break occurs), we print the subtotal, reset the subtotal to 0, and reset the variable used to compare all following records to the new id number just read in.
- Once the last record is read in and the loop ends, we will need to perform one last subtotal line prior to printing summary totals.